Structured notes: Abstract Algebra I, Module 4

Fraleigh, *A First Course in Abstract Algebra* (8th ed.), Module 4 (sections 16, 22 and 23). The syllabus names the 7th edition; items are matched by content. Numbers are the book's own. Formal version of the same content, with numbered results and the same order of items. Exercises: only the exercises the text names are reproduced, each directly after the item that names it. Figures are described with their labels; not redrawn. Factor groups and factor rings are not taught here. Sections 17 onward are not in this module.

Section 16: Isomorphism Theorems (16.1–16.10)

Printed pp. 145–148 (16.1–16.10; Figures 16.1, 16.9; Exercise 10 is named in 16.7 and printed on page 149)

Prerequisites

- Normal subgroups and factor groups G/N (Module V material; used here only as stated).
- Homomorphisms, kernels and kernel cosets (Module 2, Section 8; Module 3, Theorem 10.17).

1. First isomorphism theorem

Figure 16.1. Triangle diagram: $\phi : G \to \phi[G]$, $\gamma_K : G \to G/K$, and a dashed isomorphism $\mu : G/K \to \phi[G]$. Labels as printed; not redrawn.

THEOREM 16.2 (FIRST ISOMORPHISM THEOREM) Let $\phi : G \to G'$ be a homomorphism with kernel K, and $\gamma_K : G \to G/K$ canonical. There is a unique isomorphism $\mu : G/K \to \phi[G]$ with $\phi(x) = \mu(\gamma_K(x))$ for all $x \in G$.

2. Joins and a lemma

For $H, N \leq G$ put $HN = \{hn \mid h \in H, n \in N\}$. The join $H \vee N$ is the smallest subgroup containing HN.

LEMMA 16.3 If $N \trianglelefteq G$ and $\gamma : G \to G/N$ is canonical, then $L \mapsto \gamma[L]$ is a bijection from the normal subgroups of G containing N onto the normal subgroups of G/N.

Proof. By Theorem 13.18 and Theorem 10.17, $\gamma^{-1}[\gamma[L]] = L$ for every $L \supseteq N$, which gives injectivity. For surjectivity, $\gamma[\gamma^{-1}[H]] = H$ for every normal $H \leq G/N$, and $\gamma^{-1}[H]$ contains N. ■

LEMMA 16.4 If $N \trianglelefteq G$ and $H \leq G$, then $H \vee N = HN = NH$. If $H \trianglelefteq G$ also, then $HN \trianglelefteq G$.

Proof. For $h_1, h_2 \in H$, $n_1, n_2 \in N$, write $n_1 h_2 = h_2 n_3$ with $n_3 \in N$; then $(h_1 n_1)(h_2 n_2) = (h_1 h_2)(n_3 n_2)$, and $(hn)^{-1} = h^{-1} n_4$ with $n_4 \in N$. So $HN \leq G$; the same argument gives $NH \leq G$. If H is normal, $ghng^{-1} = (ghg^{-1})(gng^{-1}) \in HN$. ◆

3. Second isomorphism theorem

THEOREM 16.5 If $H \leq G$ and $N \trianglelefteq G$, then $(HN)/N \simeq H/(H \cap N)$.

Proof. $\phi(h) = hN$ is a homomorphism $H \to HN/N$, onto, with kernel $H \cap N$. Apply Theorem 16.2. ◆

Example 16.6. In $\mathbb{Z}^3$ with $H = \mathbb{Z}^2 \times \{0\}$ and $N = \{0\} \times \mathbb{Z}^2$: $HN = \mathbb{Z}^3$ and $H \cap N = \{0\} \times \mathbb{Z} \times \{0\}$; both quotients are $\simeq \mathbb{Z}$.

Example 16.7. In $\mathbb{Z}$ with $N = \langle n \rangle$, $H = \langle h \rangle$: $N + H = \langle \gcd(n, h) \rangle$ and $N \cap H = \langle \operatorname{lcm}(n, h) \rangle$. The second isomorphism theorem gives $\langle \gcd(n, h) \rangle / \langle n \rangle \simeq \langle h \rangle / \langle \operatorname{lcm}(n, h) \rangle$, and both are cyclic of order $n/\gcd(n, h)$. Hence $nh = \gcd(n, h) \operatorname{lcm}(n, h)$.

Exercise 10 (named in Example 16.7). Use one of the Isomorphism Theorems to prove that if $a, b \in \mathbb{Z}^+$ and $a \mid b$, then $|a\mathbb{Z}/b\mathbb{Z}| = b/a$.

4. Third isomorphism theorem

THEOREM 16.8 If $H, K \trianglelefteq G$ and $K \leq H$, then $G/H \simeq (G/K)/(H/K)$.

Proof. $\phi(gK) = gH$ is well defined since $gK \subseteq gH$; it is a homomorphism, onto, with kernel H/K. Apply Theorem 16.2: $\mu((gK)(H/K)) = gH$. ◆

Figure 16.9 (printed on page 148): an ellipse for G with the cosets of H and of K drawn inside it. No labels; not redrawn.

Example 16.10. In $\mathbb{Z}_8$ with $H = \{0, 2, 4, 6\}$ and $K = \{0, 4\}$: $G/K = \{\{0, 4\}, \{1, 5\}, \{2, 6\}, \{3, 7\}\}$, $G/H = \{\{0, 2, 4, 6\}, \{1, 3, 5, 7\}\}$, $H/K = \{\{0, 4\}, \{2, 6\}\}$, and $(G/K)/(H/K) = \{\{\{0, 4\}, \{2, 6\}\}, \{\{1, 5\}, \{3, 7\}\}\}$. The isomorphism sends $\{\{0, 4\}, \{2, 6\}\} \mapsto \{0, 4, 2, 6\}$ and $\{\{1, 5\}, \{3, 7\}\} \mapsto \{1, 5, 3, 7\}$.

Notes on this section

Book items 16.1 to 16.10 and Figures 16.1 and 16.9 are covered. Exercise 10 is printed after the sentence that names it.

Section 22: Rings and Fields (22.1–22.18)

Printed pp. 185–191 (22.1–22.18; Exercises 30, 36, 37, 38, 45, 50 named in the text)

Prerequisites

- Abelian groups; the additive group of a ring.
- Division algorithm and remainders (Module 2, Theorem 6.2).
- Plus Two recall: distributive law in school algebra (status OPEN, check against the Kerala text).

1. Rings

DEFINITION 22.1 A **ring** $\langle R, +, \cdot \rangle$: (R1) $\langle R, + \rangle$ is an abelian group; (R2) multiplication is associative; (R3) for all $a, b, c \in R$, $a(b+c) = ab + ac$ and $(a+b)c = ac + bc$.

Example 22.2. $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are rings.

Example 22.3. $M_n(R)$ is a ring for any ring R; it is noncommutative for $n \geq 2$.

Example 22.4. $F = \{f : \mathbb{R} \to \mathbb{R}\}$ with pointwise operations is a ring.

Exercise 36 (named in Example 22.4). Show that the multiplication defined on the set F of functions in Example 22.4 satisfies axioms $\mathcal{R}_2$ and $\mathcal{R}_3$ for a ring.

Example 22.5. $n\mathbb{Z}$ is a ring.

Example 22.6. $\mathbb{Z}_n$ with multiplication modulo n is a ring.

Example 22.7. Direct products of rings, by components, are rings.

2. Elementary properties

THEOREM 22.8 In any ring: (1) $0a = a0 = 0$; (2) $a(-b) = (-a)b = -(ab)$; (3) $(-a)(-b) = ab$.

Proof. (1) $a0 + a0 = a(0+0) = a0$ gives $a0 = 0$ by cancellation in $\langle R, + \rangle$. (2) $a(-b) + ab = a(-b+b) = 0$. (3) By (2) twice, $(-a)(-b) = -(-(ab)) = ab$. ◆

Exercise 30 (named after Theorem 22.8). What is wrong with the following attempt at a proof of Property 2 in Theorem 22.8?
$(-a)b = ((-1)a)b = (-1)(ab) = -(ab)$.

3. Homomorphisms and isomorphisms

DEFINITION 22.9 A map $\phi : R \to R'$ of rings is a **homomorphism** if $\phi(a+b) = \phi(a) + \phi(b)$ and $\phi(ab) = \phi(a)\phi(b)$ for all a, b. Its **kernel** is $\{a \in R \mid \phi(a) = 0'\}$.

ϕ is a group homomorphism of the additive groups, so one-to-one iff the kernel is $\{0\}$.

Example 22.10. The evaluation $\phi_a(f) = f(a)$ on F is a ring homomorphism.

Exercise 37 (named in Example 22.10). Show that the evaluation map ϕ_a of Example 22.10 is a ring homomorphism.

Example 22.11. $\mathbb{Z} \to \mathbb{Z}_n$, $a \mapsto a \bmod n$, is a ring homomorphism.

DEFINITION 22.12 A ring **isomorphism** is a bijective ring homomorphism.

Exercise 38 (named after Definition 22.12). Complete the argument outlined after Definitions 22.12 to show that isomorphism gives an equivalence relation on a collection of rings.

Example 22.13. $x \mapsto 2x$ is a group isomorphism $\mathbb{Z} \to 2\mathbb{Z}$ but not a ring homomorphism.

4. Units and fields

DEFINITION 22.14 A **commutative ring** has commutative multiplication; a **ring with unity** has a multiplicative identity 1 (**unity**).

Example 22.15. If $\gcd(r, s) = 1$, $\mathbb{Z}_{rs} \simeq \mathbb{Z}_r \times \mathbb{Z}_s$ as rings.

Exercise 45 (named in Example 22.15). Show that the multiplicative inverse of a unit in a ring with unity is unique.

DEFINITION 22.16 A **unit** of a ring with unity $1 \neq 0$ has a multiplicative inverse. A **division ring** has every nonzero element a unit. A **field** is a commutative division ring; a noncommutative one is a **strictly skew field**.

Example 22.17. The units of $\mathbb{Z}_{14}$ are $1, 3, 5, 9, 11, 13$.

Example 22.18. $\mathbb{Z}$ is not a field; $\mathbb{Q}$, $\mathbb{R}$ are.

Exercise 50 (named after Example 22.18). Show that a subset S of a ring R gives a subring of R if and only if $0 \in S$, $a - b \in S$ for all $a, b \in S$, and $ab \in S$ for all $a, b \in S$.

Notes on this section

Book items 22.1 to 22.18, with the named exercises (30, 36, 37, 38, 45, 50). Exercises 1 to 58 of the section are in the Module 4 excerpt. Factor rings (Section 30) are outside Module 4.

Section 23: Integral Domains (23.1–23.14)

Printed pp. 194–196 (23.1–23.14; Figure 23.10; Exercises 26 and 35 named in the text)

Prerequisites

- Rings, units and fields (Section 22 notes).
- gcd and relative primeness (Module 2, Section 6).
- Plus Two recall: factorising a quadratic (status OPEN, check against the Kerala text).

1. Divisors of zero and cancellation

Example 23.1. In $\mathbb{Z}_{12}$, $x^2 - 5x + 6 = (x-2)(x-3)$ and $(2)(6) = (3)(4) = 0$, so the equation has the roots $2, 3, 6, 11$.

DEFINITION 23.2 Nonzero a, b in a ring R with $ab = 0$ are **divisors of 0** (or **0 divisors**).

In $\mathbb{Z}_{12}$ the divisors of 0 are $2, 3, 4, 6, 8, 9, 10$. A unit is never a divisor of 0.

THEOREM 23.3 For $m \in \mathbb{Z}_n$, either $m = 0$; or $\gcd(m, n) = 1$ and m is a unit; or $\gcd(m, n) \neq 1$ and m is a 0 divisor.

Proof. If $d = \gcd(m, n) \neq 1$ and $m \neq 0$, then $m \cdot (n/d) = (m/d)n = 0$ in $\mathbb{Z}_n$, with m and n/d nonzero. If $\gcd(m, n) = 1$, write $an + bm = 1$, $b = nq + r$, $0 \leq r \leq n - 1$; then $rm = 1 - n(a + qm) = 1$ in $\mathbb{Z}_n$. ◆

Example 23.4. In $\mathbb{Z}_{20}$ the units are $1, 3, 7, 9, 11, 13, 17, 19$ and the 0 divisors are $2, 4, 5, 6, 8, 10, 12, 14, 15, 16, 18$.

COROLLARY 23.5 If p is prime, every nonzero element of $\mathbb{Z}_p$ is a unit; $\mathbb{Z}_p$ is a field with no divisors of 0.

Proof. Apply Theorem 23.3 with $\gcd(m, p) = 1$ for $0 < m \leq p - 1$. ■

THEOREM 23.6 Cancellation holds in a ring R if and only if R has no divisors of 0.

Proof. If $ab = 0$ with $a \neq 0$, cancellation applied to $ab = a0$ gives $b = 0$. Conversely, if $ab = ac$ with $a \neq 0$ and no divisors of 0, then $a(b - c) = 0$ gives $b = c$. ◆

In a ring with no divisors of 0, $ax = b$ with $a \neq 0$ has at most one solution. In a commutative ring b/a denotes $a^{-1}b = ba^{-1}$.

2. Integral domains

DEFINITION 23.7 An **integral domain** is a commutative ring with unity $1 \neq 0$ and no divisors of 0.

Example 23.8. $\mathbb{Z}$ and $\mathbb{Z}_p$ (p prime) are integral domains; $\mathbb{Z}_n$ is not if n is composite; a direct product $R \times S$ of nonzero rings is not, since $(r, 0)(0, s) = (0, 0)$.

Example 23.9. $\mathbb{Z}_2$ is an integral domain, but $M_2(\mathbb{Z}_2)$ has divisors of 0: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = 0$. Every field is an integral domain.

Figure 23.10. A collection of rings as overlapping regions, numbered 1 to 6, with the labels Commutative rings, Rings with unity, Integral Domains and Fields as printed. Not redrawn.

Exercise 26 (named in the text introducing Figure 23.10). Redraw Fig. 23.10 to include a subset corresponding to strictly skew fields.

THEOREM 23.11 Every finite integral domain is a field.

Proof. For $a \neq 0$ the map $x \mapsto ax$ is one-to-one by cancellation, hence onto since R is finite; so $ab = 1$ for some b. ■

3. Characteristic

DEFINITION 23.12 The least positive n with $n \cdot a = 0$ for all $a \in R$ is the **characteristic** of R; if none exists, R has characteristic 0.

Exercise 35 (named in the text after Definition 23.12). Show that the characteristic of an integral domain D must be either 0 or a prime p. [*Hint:* If the characteristic of D is mn, consider $(m \cdot 1)(n \cdot 1)$ in D.]

Example 23.13. $\mathbb{Z}_n$ has characteristic n; $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ have characteristic 0.

THEOREM 23.14 In a ring with unity, the characteristic is 0 if $n \cdot 1 \neq 0$ for all n, and otherwise the least n with $n \cdot 1 = 0$.

Proof. If $n \cdot 1 = 0$, then $n \cdot a = a(n \cdot 1) = 0$ for all a. ◆

Notes on this section

Book items 23.1 to 23.14, with Figure 23.10 and the named exercises (26 and 35). Exercises 1 to 36 of Section 23 are in the Module 4 excerpt.